

LLM-Driven Usefulness Judgment for Web Search Evaluation

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Motivation

- 相关性（relevance）：指的是文档与用户查询之间的匹配程度，通常基于Cranfield 范式。基于单条搜索结果来评估文档的相关性。
- 有用性（usefulness）：则更进一步，它衡量的是文档对用户完成其背后搜索任务的帮助程度。不仅关注内容是否相关，还关注信息是否真正解决了用户的问题。往往需要更丰富的上下文数据。

Motivation

- 尽管“相关性”长期以来一直是主导评估指标，但越来越多研究者认为它在交互式信息检索中无法充分反映系统的实际有效性。
- 本研究的动机是评估大语言模型是否能够超越传统基于相关性的评估方法，准确衡量文档的有用性，从而解决当前信息检索评估方法中的关键局限。

Motivation

- 然而，手动标注有用性标签的成本高、难度大。本文的目标是探索如何使用 LLM 结合显式与隐式用户行为信号，自动生成有用性标签，这些标签融合了搜索会话、任务、查询和文档层面的用户行为指标，力图更准确地捕捉用户对搜索系统质量的主观感知。

Overview

- 本文提出的方法包含以下四个步骤：

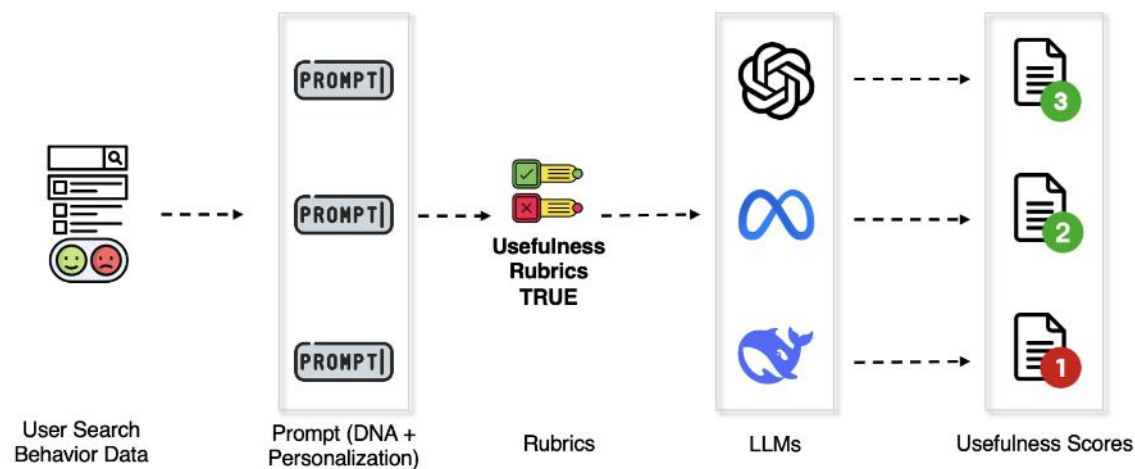


Figure 1: Usefulness label generation with LLMs.

Search Behavior Dataset

- THUIR KDD'19: 任务驱动、结构化较强，包含更长的搜索会话
- TianGong-QRef: 短会话场景，且不明确区分任务

Dataset	#Tasks	#Participants	#Sessions	#Queries	#Clicks	#Data Points
THUIR-KDD'19	9	50	447	735	1,431	3,041
QRef	—	42	2,024	4,809	7,126	7,380

Table 1: Statistics of user behavior logs.

Measure	Description
User	Participants in search task
Task	Description of what user tries to accomplish using the search engine
Query	User-submitted query
Query Position	Sequential order of the query in a task
Clicked Documents	User-clicked URL
Title and Summary	Title and summary of clicked URL in SERP
Rank	Rank of the clicked document in SERP
Task and Query Relevance	Relevance of clicked documents to task and query labeled (0-3)
Query Satisfaction	User satisfaction on a query (1-5)
Session Satisfaction	User satisfaction on a search session (1-5)
Usefulness	Usefulness feedback on a document (0-3)
URL CTR	Percentage of clicks a document receives within a task
URL Dwell Time	Time spent on a clicked document
Query Dwell Time	Time spent on a query
Task Dwell Time	Time spent on a task
Average Query Dwell Time	Average time spent on a query

Table 6: Overview of dataset measures and its descriptions.

Search Behavior Dataset

- Baseline Data: 将每条数据（无论其是否属于同一个会话）单独视为一个输入批次，称为基线数据baseline data。采用 point-wise 的方式建立实验基线，并通过评分准则推理（rubric-based reasoning）与微调技术进一步提升性能。在 THUIR-KDD'19 中约有 3,046 条数据点，在 QRef 数据集中约有 7,380 条。
- Personalized Session Data: 每个用户任务下的所有查询及其点击文档的相关指标整合为一个批次，称为session data。因为在实际应用中，判断一个文档是否有时，整个搜索会话的历史上下文非常关键。从 THUIR-KDD'19 中得到约 447 个会话，QRef 中得到约 2,024 个会话。

Prompt Techniques

- LLM 输出对提示方式非常敏感，不同提示结构与评估指标容易引发输出不一致的问题。本文采用 zero-shot DNA 提示结构，该结构由三个主要部分组成：描述部分（Descriptive）、叙述部分（Narrative）、维度部分（Aspects）
- 描述与叙述部分为 LLM 提供详细的搜索行为数据，包括用户、任务、查询、点击文档等。维度部分采用 Chain-of-Thought（CoT）推理结构，帮助 LLM 将标签判断任务分解为多个推理步骤，增强模型对查询、任务与交互之间关系的理解。
DNA+CoT（用于基线数据） DNA+CoT+Personalized（用于会话数据） 增强版本：
DNA+TRUE（将 CoT 部分替换为 TRUE（分别对应有用性等级 0, 1, 2, 3））

TRUE (Task-aware Rubric-based Usefulness Evaluation)

通过迭代方式构建了结构化评分准则：分析用户行为指标，为每个有用性标签定义清晰的评分标准，从而实现基于推理驱动的评估。

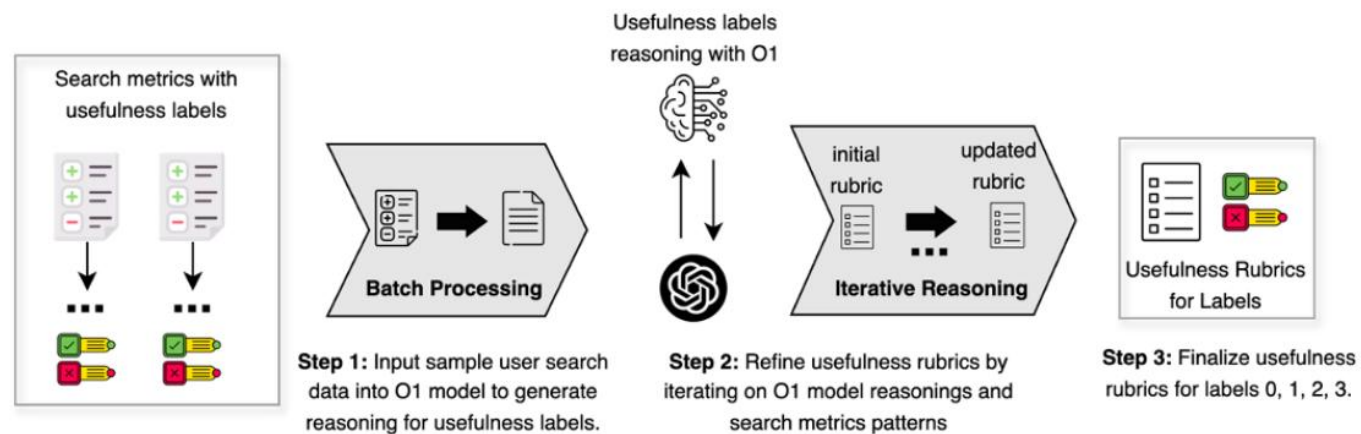


Figure 2: TRUE formulation.

TRUE

```
**Task:**
You are analyzing the usefulness scores of clicked documents of users for each task session based on the following information:
-- User Features --
Each clicked document has been provided a usefulness score in the 'usefulness score' column on an integer scale of 0 to 3 with the following meanings:
3 = Very Useful, very helpful for this query
2 = Fairly Useful, fairly helpful for this query
1 = Somewhat Useful, maybe partly helpful but might contain other irrelevant content
0 = Not Useful at all, should never be shown for this query

You must analyze each user's each task session and provide reasoning for the usefulness score of each clicked document (URL) based on all the information provided. Consider the relationship between relevance, usefulness, satisfaction and user behavior metrics like dwell time, CTR, query position, and rank.
```

Figure 6: Reasoning prompt for creating TRUE.

对应第一步，引导 LLM 分析具体点击数据背后的行为模式，从而推理某个文档为什么被打成 0、1、2 或 3 分。

评分细则-->

----- RUBRICS -----
Score 3: Very Useful - Relevance Criteria: - Task Relevance: 2-3, Query Relevance: 2-3. - Satisfaction Criteria: - Task and Query Satisfaction: 4-5. - User Behavior Metrics: - Dwell Time: Above average, CTR: High. - Title & Summary Alignment: - High Alignment (>0.5) with Query and Task. - Exceptional Cases: - High satisfaction supports high usefulness even with low relevance. - High dwell time supports high usefulness even with low CTR and low relevance. - Despite low alignment, some contents provide unexpected useful information.
Score 2: Fairly Useful - Relevance Criteria: - Task Relevance: 1-2, Query Relevance: 1-2. - Satisfaction Criteria: - Task and Query Satisfaction: 3-4. - User Behavior Metrics: - Dwell Time: Around average, CTR: Medium. - Title & Summary Alignment: - Moderate Alignment (0.3-0.5) with Query or Task. - Exceptional Cases: - Users found content useful but lacking depth leading to high relevant but <u>fairly useful</u> cases. - Moderate engagement supports moderate usefulness when content is relevant but incomplete.
Score 1: Somewhat Useful - Relevance Criteria: - Task Relevance: 1, Query Relevance: 1. - Satisfaction Criteria: - Task and Query Satisfaction: 2-3. - User Behavior Metrics: - Dwell Time: Below average, CTR: Low. - Title & Summary Alignment: - Low Alignment (<0.3) with Query or Task. - Exceptional Cases: - Low satisfaction supports low usefulness even with high relevance. - Dwell Time: Below average, CTR: Low means users clicked expecting relevance but left quickly, indicating lack of depth and low usefulness. - Misleading title-query match leads to high relevance and low usefulness cases.
Score 0: Not Useful - Relevance Criteria: - Task Relevance: 0, Query Relevance: 0. - Satisfaction Criteria: - Task and Query Satisfaction: 1-2. - User Behavior Metrics: - Dwell Time: Minimal, CTR: Very Low. - Title & Summary Alignment: - No Alignment (<0.2) with Query or Task. - Low alignment with high usefulness indicates LR-HU. - Exceptional Cases: - Very low satisfaction confirms low relevance and low usefulness cases. - Dwell Time: Minimal, CTR: Very Low means users abandoned the page quickly due to irrelevance leading to low usefulness. - Low dwell time with high CTR indicates low usefulness even with high relevance. - No Alignment (<0.2) with Query or Task means content is completely off-topic, reinforcing low relevance and low usefulness cases.

Figure 9: Rubric for THUIR-KDD'19 dataset.

TRUE和CoT对比

```
**Task:**
You are a search quality rater evaluating the usefulness of clicked documents based on
the following information:
  a. ID: Unique ID for each row.
  b. User: User ID
  c. Task ID: Unique ID for each task.
  d. Task Description: A clear explanation of what the user tries to accomplish using
the search engine.
  e. Query: User submitted query.
  f. Query Position: The sequential order of the query within the task session,
indicating when it was submitted relative to other queries.
  g. Clicked Documents: User clicked URL.
  h. Title: Title of the clicked documents.
  i. Summary: Brief description of the clicked documents on the web page.
  j. Rank: Rank of clicked documents on the SERP labeled as 0(high)-30(low).
  k. Task Relevance: Relevance of clicked documents to the user's task labeled as 0
(low)-3 (high).
  l. Query Relevance: Relevance of clicked documents to the user's query labeled as 0
(low)-3 (high).
  m. Task Satisfaction: User's satisfaction on overall task labeled as 1(low)-5(high).
  n. Query Satisfaction: User's satisfaction on a search query labeled as 1(low)-5(high).
  o. CTR: The percentage of clicks a document (URL) receives out of total interactions
within a specific task and query.
  p. URL Dwell Time (ms): Time spent on a clicked document.
  q. Average URL Dwell Time (ms): Average time a user spends across all clicked document
on individual task.
  r. Task Dwell Time (ms): Time spent on a specific task across all queries and clicked
documents.
  s. Query Dwell Time (ms): Time spent on a specific query

You must analyze each task session by considering all the information given above and for
each clicked document (URL), provide a usefulness score on an integer scale of 0 to 3
with the following meanings:
3 = Very Useful, very helpful for this query
2 = Fairly Useful, fairly helpful for this query
1 = Somewhat Useful, maybe partly helpful but might contain other irrelevant content
0 = Not Useful at all, should never be shown for this query

**Instructions:**
For each clicked documents <URL> under a specific task <TASK_ID> and user <USER>, use the
following rubrics while assigning usefulness score:

-- RUBRICS --

**Output Format:**
Take a step back and reflect carefully on how best to solve your task. Understand the
entire rubrics first and then decide on the usefulness scores analyzing the metrics.
Directly output the usefulness score for unique ID <ID> as an integer value in the
following JSON format:
{"<id_1>": <usefulness_1>, ..., "<id_n>": <usefulness_n>}

DO NOT PROVIDE ADDITIONAL TEXT, REASONING, EXAMPLE, OR CODE. GENERATE USEFULNESS SCORE IN
COORECT OUTPUT FORMAT FOR ALL UNIQUE ID.
```

Figure 8: Rubric (DNA+TRUE) prompt for THUIR-KDD'19.

```
You are a search quality rater evaluating the usefulness of clicked documents on a web page.
You have been given the following information:
  a. ID: Unique ID for each row.
  b. User: User ID
  c. Task ID: Unique ID for each task.
  d. Task Description: A clear explanation of what the user tries to accomplish using
the search engine.
  e. Query: User submitted query.
  f. Query Position: The sequential order of the query within the task session, indicating
when it was submitted relative to other queries.
  g. Clicked Documents: User clicked URL.
  h. Title: Title of the clicked documents.
  i. Summary: Brief description of the clicked documents on the web page.
  j. Rank: Rank of clicked documents on the SERP labeled as 0(high)-30(low).
  k. Task Relevance: Relevance of clicked documents to the user's task labeled as 0(low)-3
(high).
  l. Query Relevance: Relevance of clicked documents to the user's query labeled as 0 |
(low)-3 (high).
  m. Task Satisfaction: User's satisfaction on overall task labeled as 1(low)-5(high).
  n. Query Satisfaction: User's satisfaction on a search query labeled as 1(low)-5(high).
  o. CTR: The percentage of clicks a document (URL) receives out of total interactions
within a specific task and query.
  p. URL Dwell Time (ms): Time spent on a clicked document.
  q. Average URL Dwell Time (ms): Average time a user spends across all clicked document
on individual task.
  r. Task Dwell Time (ms): Time spent on a specific task across all queries and clicked
documents.
  s. Query Dwell Time (ms): Time spent on a specific query

You must analyze each task session by considering all the information given above and for
each clicked document (URL), provide a usefulness score on an integer scale of 0 to 3 with
the following meanings:
3 = Very Useful, very helpful for this query
2 = Fairly Useful, fairly helpful for this query
1 = Somewhat Useful, maybe partly helpful but might contain other irrelevant content
0 = Not Useful at all, should never be shown for this query

Important Instructions:
Consider all the attributes above while deciding on a usefulness score. If certain attributes
are unavailable, rely on the available ones to decide. Assign category 1 if the clicked
document is somewhat useful to the task and query but not completely, category 2 if the
clicked document presents something very important related to the task and query but also
has some extra information, and category 3 if the clicked document only and entirely refers
to the task and document. If none of the above satisfies give it category 0.

You will be provided data in the form of:
--JSON Format of Baseline or Session Data--

For each clicked documents <URL> under a specific task <TASK_ID> and user <USER>, split
this problem into following steps:
  a. Consider all the attributes and relative importance of each and decide on a final
score <usefulness_i>. Final score must be an integer value only.
  b. Prioritize all user metrics like CTR, URL dwell time, query dwell time and session
dwell time as indicators of usefulness.
  c. Consider user's intent of the search and task session ensuring they align with the
title and summary.
  d. Consider the rank of URLs relation to the user's query and task description.
  e. Consider task relevance and query relevance which indicate how well the result aligns
with the user's overall goal.
  f. Consider user's query satisfaction score and session satisfaction score, which reflect
their satisfaction with the result.

Directly output the usefulness score for each unique row <ID> as an integer value in the
following JSON format:
{"<id_1>": <usefulness_1>, ..., "<id_n>": <usefulness_n>}.
Do not provide additional reasoning, examples, or code.
```

Figure 7: Baseline (DNA+CoT) and session (DNA+CoT+Personalization) prompts.

Experimental Setup

- 实验共分为三部分：1.使用 baseline data 建立基准；2.使用 session data 和 TRUE 准则进行评估；3.在 baseline data 上引入 TRUE 并进行微调，以进一步提升模型性能。
- 提示均采用zero-shot
- 模型包括：GPT-3.5 Turbo、GPT-4o、LLaMA 3.3 70B-Instruct、3.2 3B-Instruct、3.1 8B-Instruct
- 以THUIR-KDD'19 和 QRef 中的人工标注有用性标签（0-3）作为ground truth
- Spearman 等级相关系数作为评价指标

Baseline, Session and Rubrics Data

- 基线（Baseline, B）：DNA + CoT 提示结构
- 会话感知（Session-aware, S）：DNA + CoT + Personalized
- 评分规则方法（Rubric, R）：DNA + TRUE 方法

LLM	Method	THUIR-KDD'19			QRef		
		Overall	Task	Query	Overall	Session	Query
LLaMA 3.3 70B	B _{DNA+CoT}	0.38	0.34	0.30	0.33	0.38	0.44
	S _{DNA+CoT+Personalization}	0.37	0.32	0.28	0.37	0.41	0.42
	R _{DNA+TRUE (Ours)}	0.48	0.48	0.36	0.38	0.43	0.47
	$\Delta_{R \text{ vs } B}$	$\uparrow 0.10$	$\uparrow 0.14$	$\uparrow 0.06$	$\uparrow 0.05$	$\uparrow 0.05$	$\uparrow 0.03$
LLaMA 3.1 8B	B _{DNA+CoT}	0.31	0.26	0.20	0.25	0.29	0.39
	S _{DNA+CoT+Personalization}	0.31	0.27	0.27	0.27	0.32	0.39
	R _{DNA+TRUE (Ours)}	0.37	0.37	0.24	0.26	0.25	0.33
	$\Delta_{R \text{ vs } B}$	$\uparrow 0.06$	$\uparrow 0.11$	$\uparrow 0.04$	$\uparrow 0.01$	$\downarrow 0.04$	$\downarrow 0.06$
LLaMA 3.2 3B	B _{DNA+CoT}	0.19	0.12	0.06	0.13	0.23	0.32
	S _{DNA+CoT+Personalization}	0.22	0.17	0.21	0.30	0.37	0.46
	R _{DNA+TRUE (Ours)}	0.28	0.28	0.22	0.05	0.03	0.07
	$\Delta_{R \text{ vs } B}$	$\uparrow 0.09$	$\uparrow 0.16$	$\uparrow 0.16$	$\downarrow 0.08$	$\downarrow 0.20$	$\downarrow 0.25$
GPT-4o-mini	B _{DNA+CoT}	0.35	0.30	0.28	0.29	0.35	0.44
	S _{DNA+CoT+Personalization}	0.36	0.30	0.25	0.37	0.41	0.48
	R _{DNA+TRUE (Ours)}	0.43	0.36	0.32	0.35	0.32	0.35
	$\Delta_{R \text{ vs } B}$	$\uparrow 0.08$	$\uparrow 0.06$	$\uparrow 0.04$	$\uparrow 0.06$	$\downarrow 0.03$	$\downarrow 0.09$
GPT-3.5 Turbo	B _{DNA+CoT}	0.37	0.31	0.26	0.21	0.34	0.42
	S _{DNA+CoT+Personalization}	0.33	0.28	0.26	0.37	0.45	0.47
	R _{DNA+TRUE (Ours)}	0.37	0.33	0.31	0.09	0.30	0.16
	$\Delta_{R \text{ vs } B}$	$\uparrow 0.00$	$\uparrow 0.02$	$\uparrow 0.05$	$\downarrow 0.12$	$\downarrow 0.04$	$\downarrow 0.26$

Table 2: Spearman’s rank correlation comparison across models and prompting methods on THUIR-KDD’19 and QRef dataset.

Baseline, Session and Rubrics Data

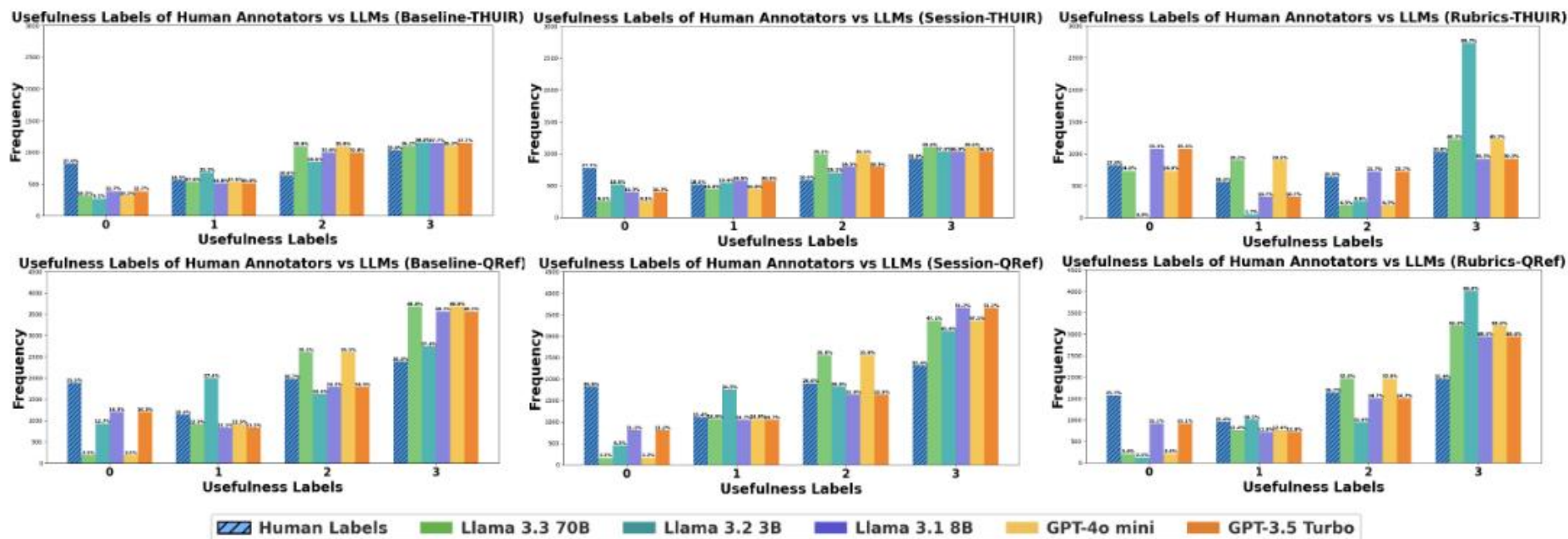


Figure 3: Distribution of usefulness labels (human vs LLMs).

各方法下标签的分布情况

Relevance vs Usefulness

- 将 THUIR-KDD'19 数据集中的样本按以下四种组合进行划分：

HR-HU（高相关，高有用）、HR-LU（高相关，低有用）、LR-HU（低相关，高有用）、LR-LU（低相关，低有用）

用三个方法（B, S, R）在五个 LLM 上分别测量其 Spearman 等级相关性

LLM	HR-HU			HR-LU			LR-HU			LR-LU		
	B	S	R	B	S	R	B	S	R	B	S	R
GPT-3.5 Turbo	0.10	0.12	0.14	0.02	0.09	0.23	0.17	0.04	-0.01	0.07	0.05	0.10
GPT-4o-mini	0.12	0.13	0.16	-0.13	0.11	0.16	0.10	0.05	-0.07	0.05	0.13	0.08
LLaMA 3.1 8B	0.09	0.11	0.12	0.05	0.03	0.21	0.12	0.04	-0.01	0.04	0.11	0.12
LLaMA 3.2 3B	0.06	0.06	0.09	0.04	0.05	0.11	0.19	0.05	0.12	0.03	0.03	0.09
LLaMA 3.3 70B	0.16	0.14	0.21	0.13	0.08	0.30	0.01	0.06	-0.02	0.11	0.11	0.19

Table 3: Comparison of spearman's correlation (THUIR-KDD'19) across cases: HR-HU, HR-LU, LR-HU, and LR-LU.

Ablation Study

- 在生成文档有用性标签时，哪些输入特征对 LLM 最为关键。将特征划分为五类：查询特征（Q）、文档特征（D）、相关性指标（R）、满意度指标（S）、用户行为指标（U）
- 固定Q 和 D，只对R、S和U这三类特征进行组合变化，以观察它们对模型效果的影响。

Features	GPT-4o-mini	LLaMA 3.3 70B
U_{R+S+U}	0.36	0.37
U_{R+S}	0.35	0.35
U_{R+U}	0.34	0.33
U_{S+U}	0.34	0.31
U_R	0.32	0.33
U_S	0.27	0.29
U_U	0.27	0.28

Table 4: Ablation study of features in GPT-4o-mini and LLaMA 3.3 70B with session data.

Finetuning

- 对LLaMA-3.1-8B-Instruct 进行微调
- 设计了一个紧凑的 zero-shot DNA+TRUE 提示模版，结合了 DNA 提示结构和 TRUE 评分框架。基于 THUIR-KDD'19 和 QRef 数据集构建了一个定制数据集用于微调。
- 使用 LoRA 技术
- 应用 4-bit quantization and 8-rank adapters
- 设置最大上下文长度为 2048 tokens
- 模型在 2 个 epoch 内收敛

Finetuning

Metric	THUIR		QRef	
	PT	FT	PT	FT
MSE	0.19	0.20	0.41	0.42
MAE	0.19	0.20	0.41	0.42
Acc	0.81	0.80	0.59	0.58
F1	0.64	0.62	0.41	0.40

Table 5: Comparison of LLM performance pre-trained (PT) vs fine-tuned (FT) (binary labels.)

Discussion

- LLM生成的有用性标签依赖于模型参数和规模大小。
- LLM生成的有用性标签也受到数据集类型与提示设计方法的影响。
- 对于具有丰富用户行为和上下文的长会话，TRUE 方法效果最好；
- 对于更短的会话，采用 session prompting（会话感知提示）能更好地帮助模型理解用户的搜索行为和语境关系。

Conclusion

- 优势：充分利用了 LLM 对上下文关系和语义的理解能力，为用户提供更符合实际感知的评估方式。
- 局限：有用性预测的性能仍显著低于相关性判断；缺乏统一的、有标准的数据集用于有用性评估